

Complete Scalar Actions and Identifiability Boundaries in the ACS Framework

Brad Wallace
Independent Researcher

24 September 2026

Abstract

We consolidate the action-to-spectrum audit of the specified minimal Pati–Salam scalar sector of the Asymmetric Codependent Systems framework. The calculation retains 68 real scalar coordinates, 17 real quartic coefficients, four quadratic coefficients and the trace kinetic metric. Six quartic directions are invisible on the neutral background family. Explicit bounded potentials with identical neutral data nevertheless have different transverse stability, and the usual two-trace restriction is not generically closed under one-loop running. A homogeneous family of complete actions changes all physical masses without changing dimensionless labels; the result extends to arbitrary complex flavor matrices and the stated one-loop potential. Independent finite-running calculations retain boundary sensitivity. These results establish a usable forward calculation and precise limits on what it identifies. They do not select a physical vacuum, a unique mass scale or measured flavor data. Exact identities, numerical qualifications, source errors and unresolved physical inputs are reported separately.

1 Introduction

An algebraic representation specifies allowed interactions and charge labels. A physical prediction also requires normalized coefficients, a vacuum and a prescription relating the model to measured quantities. The purpose of this paper is to determine which of these steps follow from the supplied ACS action and which remain additional data. It is a consolidation and correction of the repository’s calculations, not a claim that an unrestricted theory of matter has been completed.

Our organization follows conventional scalar-potential and matching studies: define the fields and conventions, analyze the full potential, distinguish local stability from boundedness and global selection, and state the domain of the matching calculation. Orbit-space restrictions are relevant to boundedness questions [1]. Finite matching also requires attention to mixed heavy/light contributions and normalization [2, 3]. These references provide methods and context, not evidence for an ACS parameter-selection law.

2 Fields, action and normalization

The gauge group is $SU(4) \times SU(2)_L \times SU(2)_R$, with $\Phi \sim (1, 2, 2)$ and $\Delta \sim (\overline{10}, 1, 3)$. We represent Φ by a complex 2×2 matrix and Δ by three complex symmetric 4×4 matrices D_a . Thus Φ supplies eight and Δ supplies sixty real coordinates. Define

$$N_\Phi = \text{Tr } \Phi^\dagger \Phi, \quad d_\Phi = \det \Phi, \quad N_\Delta = \sum_{a=1}^3 \text{Tr } D_a^\dagger D_a. \quad (1)$$

The potential in the inherited invariant basis is

$$V(x) = \Omega + \sum_{i=1}^4 m_i I_{2i}(x) + \sum_{j=0}^{16} \lambda_j I_{4j}(x), \quad (2)$$

where m_i is a mass-squared coefficient, not a mass. The quadratic basis is $(N_\Phi, \text{Re } d_\Phi, \text{Im } d_\Phi, N_\Delta)$. The real quartic basis is summarized in Table 1.

Indices	Invariants, in the archived normalization
0–5	$N_\Phi^2, d_\Phi ^2, \text{Re } d_\Phi^2, \text{Im } d_\Phi^2, N_\Phi \text{Re } d_\Phi, N_\Phi \text{Im } d_\Phi$
6–10	$\ P_{35,0}(D \otimes D)\ ^2, \ P_{35,2}(D \otimes D)\ ^2, \ P_{20,0}(D \otimes D)\ ^2, \ P_{20,2}(D \otimes D)\ ^2, \ P_{45,1}(D \otimes D)\ ^2$
11–12	$\text{Re } H_\Delta, \text{Im } H_\Delta$
13–16	$N_\Phi N_\Delta, \text{Re } d_\Phi N_\Delta, \text{Im } d_\Phi N_\Delta, J_\Phi \cdot J_\Delta$

Table 1: The first projector label denotes the color-representation dimension and the second the weak spin. The five norms sum to N_Δ^2 . The phase-sensitive invariant H_Δ and the weak-vector contraction use the exact component definitions in the supplied basis code; their coefficients cannot be compared across normalizations without that map.

The trace kinetic terms give $\mathcal{L}_{\text{kin}} = \frac{1}{2} \partial_\mu x^T G \partial^\mu x$, with $G = 2$ on Φ and diagonal D_a entries and $G = 4$ on off-diagonal D_a entries. If H is the raw real-coordinate Hessian, masses solve

$$Hv = m^2 Gv, \quad H_c = G^{-1/2} H G^{-1/2}. \quad (3)$$

A raw Hessian eigenvalue is consequently not automatically a physical mass. For one radial coordinate with kinetic coefficient Z and potential $c_4 r^4$, the convention $h = \sqrt{Z}r$ gives

$$V = \frac{\lambda_{\text{can}}}{4} h^4, \quad \lambda_{\text{can}} = \frac{4c_4}{Z^2}. \quad (4)$$

Consistent changes of field coordinates leave this ratio unchanged. Multiplying the entire action by an unspecified constant C instead changes the canonically normalized quartic to λ_{can}/C .

3 What a neutral restriction cannot determine

Consider

$$\Phi = \text{diag}(a, be^{i\alpha}), \quad (D_1, D_2, D_3) = \frac{dE_{44}}{\sqrt{2}}(1, -i, 0). \quad (5)$$

For nonzero unequal a, b and nonzero d , the unbroken group is $SU(3)_c \times U(1)_{\text{em}}$. The 21 gauge generators have twelve independent orbit directions, leaving 56 physical real scalar directions. The entire neutral restriction has quartic rank eleven. The missing six directions are $I_6, I_8, I_9, I_{10}, \text{Re } H_\Delta, \text{Im } H_\Delta$. The first four affect transverse masses. At the rank-one color background, H_Δ has zero value, gradient and Hessian but nonzero cubic vertices. For example, a specified diagonal perturbation gives $H_\Delta = 3ABCd/\sqrt{2}$ in the archived normalization.

3.1 An explicit stability counterfamily

Take the bounded family

$$V_4 = N_\Phi^2 + |d_\Phi|^2 + N_\Delta^2 + k(I_6 + I_8 + I_9 + I_{10}) + \frac{\text{Re } H_\Delta}{100} + \frac{\text{Im } H_\Delta}{50} + \frac{N_\Phi N_\Delta}{10} + \frac{J_\Phi \cdot J_\Delta}{50}, \quad (6)$$

$$V_2 = -\frac{2743}{7200}N_\Phi - \frac{41}{240}\text{Re } d_\Phi - \frac{1259}{625}N_\Delta. \quad (7)$$

The background $a = 1/3, b = 1/5, d = 1, \alpha = 0$ is stationary for each tested k . The neutral potential and its four physical neutral masses are identical, while Table 2 shows different full spectra. The boundedness certificate and exact sparse-polynomial calculation are retained in the canonical-vacuum evidence package. This is a deliberately chosen witness family, not an observed particle fit.

k	Positive physical modes	Flat physical modes	Negative physical modes
1/5	56	0	0
0	44	12	0
-1/5	6	0	50

Table 2: Twelve gauge Goldstone directions are excluded from every row. The neutral data alone do not distinguish the rows.

There is an exact independent negative-direction witness. The canonically normalized real part of $D_{3,00}$ has

$$m^2 = \frac{5}{3}k + \frac{4}{5625}. \quad (8)$$

It is negative at $k = -1/5$. At $k = 1/5$, the minimum physical scalar mass-squared is approximately 0.04321111. Positivity of this local spectrum does not prove that the point is the global minimum or that quantum decay is absent.

4 Radiative closure and matching

Let $B = 32\pi^2 d/d \log \mu$. The additive gauge source for the five Hermitian Δ quartics is

$$B \begin{pmatrix} \lambda_6 \\ \lambda_7 \\ \lambda_8 \\ \lambda_9 \\ \lambda_{10} \end{pmatrix}_g = g_4^4 \begin{pmatrix} 45 \\ 45 \\ 9 \\ 9 \\ -3 \end{pmatrix} + g_4^2 g_R^2 \begin{pmatrix} -72 \\ 36 \\ 72 \\ -36 \\ 12 \end{pmatrix} + g_R^4 \begin{pmatrix} 36 \\ 18 \\ 36 \\ 18 \\ 6 \end{pmatrix}. \quad (9)$$

These terms were checked monomial by monomial against the complete gauge Gram, with scalar and Yukawa contributions treated separately. The radial source, gauge sources and a further scalar-loop contraction require the full five-dimensional Hermitian space for generic independent gauge couplings. This does not preclude a future derived relation among boundary values.

The two holomorphic coefficients satisfy, at this order,

$$B\lambda_j = \left[\frac{80}{3}\lambda_8 + \frac{64}{3}\lambda_9 - 108g_4^2 - 48g_R^2 + 8\text{Tr}(F^\dagger F) \right] \lambda_j, \quad j = 11, 12. \quad (10)$$

Their simultaneous zero surface is thus preserved by these one-loop equations. It is a boundary choice, and the associated classical global charge has weak mixed anomalies. A perturbative zero surface is not a proof of exact quantum charge protection.

For illustration, both natural readings of the historical two-trace Δ potential require $\lambda_6 = \lambda_7$. At the source point $\rho_1 = 8/9, \rho_2 = 0$, equal gauge couplings $4/3$ and zero Yukawas, the complete equations give $B(\lambda_6 - \lambda_7) = -2560/9$. The reduced ansatz therefore fails a direct closure test.

Tree elimination is meaningful after an action and branch are specified. For a radial deviation y with terms $\lambda_7 y^2 + K X y$, eliminating y gives the correction $-K^2 X^2 / (4\lambda_7)$. The retained finite one-loop study additionally includes mixed heavy/light effects, infrared subtraction and finite kinetic normalization on its declared stable branch. It was checked against 80-digit limits. It assumes a particular light doublet, portal structure and real diagonal flavor; general noncommuting-flavor, mixed-quartic finite matching is not asserted.

5 A complete-action scale family

Proposition 1. *For the flat-space action with potential (2), fix the dimensionless coefficients and positive kinetic metric. Simultaneously replace*

$$x \mapsto ax, \quad m_i \mapsto a^2 m_i, \quad \Omega \mapsto a^4 \Omega, \quad \mu \mapsto a\mu, \quad a > 0. \quad (11)$$

Stationary configurations map to stationary configurations, physical mass-squared eigenvalues scale by a^2 , and their signs and ratios are unchanged. The same scaling holds for gauge masses and fermion squared singular masses, including complex noncommuting Yukawa matrices.

Proof. Each I_2 is homogeneous of degree two and each I_4 of degree four. Consequently $V, \nabla V, H$ scale by a^4, a^3, a^2 , respectively. Since G is fixed, the canonical Hessian has the same scaling. Infinitesimal gauge-orbit vectors are linear in x , so their mass Gram is quadratic. The Weyl mass matrix is linear in the scalar background at fixed Yukawas; its singular values scale by a . No diagonal-flavor assumption enters. $\square$

The statement compares physically different actions. It is not a coordinate symmetry of one fixed action. At a qualified stable background the one-loop potential has the form

$$V_1 = \frac{1}{64\pi^2} \sum_{\nu} n_{\nu} M_{\nu}^4 \left(\log \frac{M_{\nu}^2}{\mu^2} - c_{\nu} \right). \quad (12)$$

The logarithms are unchanged under the simultaneous transformation, so V_1 scales by a^4 . Two component tests use all 68 scalars, 21 gauge directions and a complex 48-Weyl example; normalized residuals are below 10^{-10} for the component scaling and 10^{-9} for the one-loop sum. These checks support the exact homogeneity argument; they do not promote the one-loop approximation to an all-orders result.

A fixed measured Newton constant could provide extra dimensional data, but the matter-sector matching relation would still have to be derived. Dimensional transmutation also retains an integration constant: $\Lambda = \mu \exp[-1/(2bg^2(\mu))]$ for the stated one-loop equation. The beta function alone does not select Λ in physical units.

6 Running does not supply the missing selector

A finite regular flow of a smooth ordinary differential equation is locally invertible. Focusing may reduce sensitivity without making all boundary data identical. The historical source’s one-loop SM truncation gives the three conditional pairs

$$(\lambda_{UV}, \lambda_{IR}) \simeq (0.08, 0.1685520), (0.1283001, 0.1806315), (0.18, 0.1898121). \quad (13)$$

A variational equation and independent boundary differences agree on $d\lambda_{IR}/d\lambda_{UV} = 0.2088898$ at the middle point. The comparison input is 0.1294. The middle RG output corresponds to an approximately 148 GeV tree mass proxy, not a pole-mass prediction. Mixed historical input conventions and omitted thresholds preclude a precision phenomenological interpretation.

The source’s fixed-gauge four-quartic proxy can also be resolved exactly. Write $\lambda_i = (16/9)(x, y, z, w)_i$. Its fourth equation is $4w(x + y - 3) = 0$. If $w \neq 0$, the second equation becomes $2z^2 + 2w^2 + 3 = 0$, which has no real solution. With $w = 0$, the third equation splits into $z = 0$ or $z = (3 - x - y)/2$. Exact elimination and Sturm root counts leave precisely two real solutions:

$$(\lambda_1, \lambda_2, \lambda_3, \lambda_4) \simeq (0.1811535, 0.5106919, 0, 0), (0.6447133, 0.5764245, 0, 0). \quad (14)$$

Neither gives $2\sqrt{3}/27$. These are roots of a fixed-gauge proxy, not fixed points of the complete running action. The full archived program timed out in its symbolic solve; that execution failure remains recorded. The independent real-root result does not relabel the failed replay.

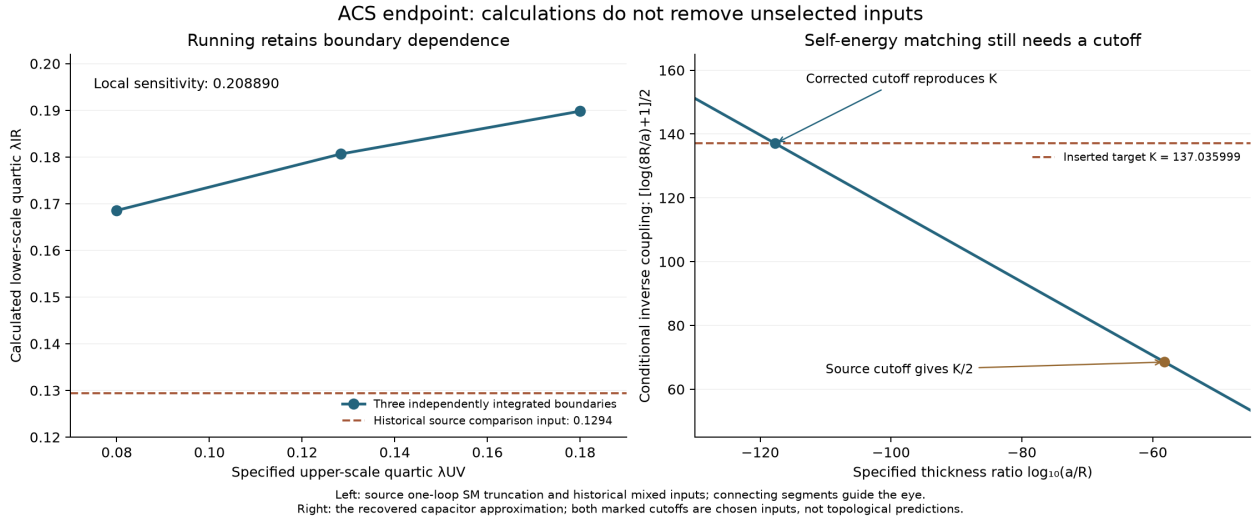


Figure 1: Left: explicit residual sensitivity to a chosen UV boundary. Right: the separate capacitor proposal, discussed in the companion carrier paper, reproduces a target only after a cutoff is specified. Both panels are conditional calculations, not empirical parameter predictions.

7 Discussion and conclusions

The completed forward calculation is useful precisely because it exposes what the neutral restriction misses. It supplies physical spectra in a specified kinetic metric, tests symmetry and radiative closure, and supports conditional matching calculations. It also supplies explicit countermodels to

uniqueness claims. A numerical minimizer cannot choose which allowed action nature uses, and a source expression matched to an inserted value does not derive that value.

The remaining physical requirements are a normalized microscopic condensate-to-field map, a coefficient and flavor selection law, and a physical vacuum and matching prescription. The present result is underdetermination within the reviewed assumptions, not a no-go theorem against a future model that adds justified premises. It neither proves RH nor identifies a universal physical origin for prime or zero statistics.

A Evidence and reproduction

The computational record is in `docs/condensate_energy/`, with the canonical, hidden-coupling, action-matching, finite-matching and endpoint subdirectories separating claims and approximations. Source snapshots, failed runs, exact polynomials and independent numerical qualifications are retained. The publication driver in `scripts/verify_publication.py` checks the portable evidence integrity and invokes the declared scientific checks; archive recovery programs requiring the original local drives are not advertised as portable. Checks are not independent experiments. The endpoint’s 95 passing checks and one incomplete source replay have different meanings and remain separate.

Acknowledgments and preparation

This consolidation used AI-assisted drafting and computational checking. The repository records the executable evidence and scope limits; no external peer-review endorsement is implied.

References

- [1] K. Kannike, “Vacuum Stability of a General Scalar Potential of a Few Fields,” *Eur. Phys. J. C* **76**, 324 (2016), arXiv:1603.02680; corrected version 3.
- [2] M. Jiang, N. Craig, Y.-Y. Li and D. Sutherland, “Complete One-Loop Matching for a Singlet Scalar in the Standard Model EFT,” *JHEP* **02** (2019), 031, arXiv:1811.08878.
- [3] V. Gherardi, D. Marzocca and E. Venturini, “Matching scalar leptoquarks to the SMEFT at one loop,” *JHEP* **07** (2020), 225, arXiv:2003.12525; version 5.